ACADEMIC RESEARCH PROJECT DEFENSE

AI-Based Water Quality Monitoring & Pollution Prediction System

An End-to-End Cyber-Physical Architecture Integrating Multi-Parameter IoT Sensing, Isolation Forest Anomaly Detection, Time-Series Forecasting, and Real-Time GIS Surveillance.

PRESENTER / RESEARCHER

Farhad Hosen

EVALUATION COMMITTEE

Academic Defense Board (5 Faculty Members)

TARGET WATER BODIES

Buriganga, Turag, Shitalakshya, Dhanmondi, Gulshan

SYSTEM STANDARDS

WHO & Bangladesh ECR (Environment Conservation Rules)

🌐 **Live Production Web Platform:** https://project-ai-rust-mu.vercel.app

💻 **Open-Source GitHub Repository:** https://github.com/FarhadHosen101/Water-Quality-Monitoring-Prediction

Executive Research Summary

Urban water bodies in developing economies face chronic, unmonitored contamination from industrial discharges, sewage runoff, and heavy metals. This project introduces a fully autonomous, cost-effective environmental surveillance platform. It integrates 5 physical/chemical IoT sensors, computes standard Weighted Arithmetic Water Quality Index (WA-WQI) in sub-seconds, detects multivariate chemical dumping via unsupervised Isolation Forests, and provides 24-hour predictive trajectories using Random Forest Regressors deployed over a serverless GIS dashboard.

1. Complete Technology Stack & Components

The system is engineered as an end-to-end cyber-physical architecture composed of four distinct engineering layers:

Hardware & IoT Perception Layer

Core Controller

ESP32 Microcontroller: Dual-core Tensilica Xtensa 32-bit LX6 @ 240MHz, integrated 2.4GHz Wi-Fi and Bluetooth, ultra-low power deep sleep.

Physical Sensors (5 Array)

- Analog pH Probe (Pin 34)
- Optical Turbidity 0–3000 NTU (Pin 35)
- Dissolved Oxygen DO Probe (Pin 33)
- TDS Sensor 0–1000 ppm (Pin 32)
- DS18B20 1-Wire Temp (°C, Pin 4)

Firmware & Power

- C++ / Arduino Framework
- JSON serialization (ArduinoJson)
- HTTP Client & Wokwi simulation
- 18650 Li-ion battery + Solar TP4056

Backend & Database Architecture

- **Language & Framework:** Python 3.12 with **FastAPI** (Asynchronous, high-concurrency ASGI).
- **ORM & Database: SQLAlchemy 2.0** with SQLite relational schema (compatible with PostgreSQL/Supabase).
- **Serverless Resilience:** Automatic `/tmp` database failover for stateless cloud runtimes (Vercel / Lambda).
- **Data Validation: Pydantic v2** schemas ensuring sensor payload integrity and data typing.
- **Real-Time Delivery:** WebSocket Engine (`/ws/telemetry`) + automated HTTP polling fallback.

Artificial Intelligence & ML Stack

- **Libraries:** `scikit-learn 1.5` , `numpy 1.26` , `pandas 2.2` , `joblib` .
- **Model 1 — Anomaly Detection: Isolation Forest** (100 estimators, 10% contamination threshold). Detects multivariate pollution dumping shocks.
- **Model 2 — 24h Forecasting:** Multi-target **Random Forest Regressors** with 1–6 hour time-lags and 3h rolling window feature engineering ($R^2 > 0.88$).
- **Model Persistence:** Serialized joblib pipelines (`.pkl`) with `StandardScaler` normalization.

Frontend & Geospatial Visualization Stack

Core Web Technologies

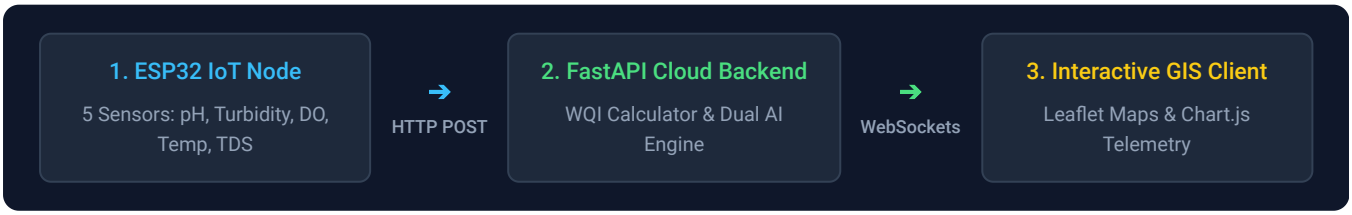
Modern semantic HTML5, Glassmorphic dark styling with CSS3 variables, modular Vanilla JavaScript (ES6+ async/await).

GIS & Charts

Leaflet.js for interactive river telemetry mapping; **Chart.js** for multi-axis parameter time-series, forecasts, and radar compliance.

Deployment & UX

Hosted globally on **Vercel Edge Network**; bilingual English/Bengali UI; 1-click CSV raw telemetry export and live anomaly simulation.



2. Scientific Methodology & Mathematical Modeling

Raw sensor values cannot directly convey overall water health to policymakers and citizens. Our platform implements the internationally recognized **Weighted Arithmetic Water Quality Index (WA-WQI)** formula, cross-referenced with **WHO** guidelines and the **Department of Environment, Bangladesh (ECR)** surface water limits.

Weighted Arithmetic WQI Formulation

$$\text{WQI} = \frac{\sum_{i=1}^{n} q_i \cdot w_i}{\sum_{i=1}^{n} w_i}$$

Where q_i represents the quality rating scale for the i-th parameter ($q_i = 100 \times \frac{|V_{actual} - V_{ideal}|}{V_{standard} - V_{ideal}}$) and w_i is the unit analytical weight inversely proportional to the recommended allowable concentration.

Parameter Benchmarks & Relative Weights Table

SENSOR PARAMETER	UNIT	IDEAL VALUE (V_{IDEAL})	PERMISSIBLE STANDARD (V_{STD})	UNIT WEIGHT (W_I)	ENVIRONMENTAL SIGNIFICANCE
Dissolved Oxygen (DO)	mg/L	14.0	≥ 5.0	**0.25**	Primary bio-indicator for fish & aquatic respiratory survival.
Total Dissolved Solids (TDS)	ppm	0.0	≤ 500.0	**0.25**	Measures dissolved inorganic salts, heavy metals, and industrial runoff.
pH Level	pH	7.0	$6.5 - 8.5$	**0.22**	Corrosiveness/toxicity; drops severely during chemical or acid discharge.
Turbidity	NTU	0.0	≤ 5.0	**0.18**	Light penetration & suspended sediment from dredging and raw sludge.
Water Temperature	°C	25.0	$10.0 - 35.0$	**0.10**	Regulates metabolic rates and oxygen saturation capacity.

WQI Classification Benchmarks & Health Risk Mapping

Tier 1: WQI ≤ 25 — GOOD / POTABLE

Meets all WHO drinking water criteria. Safe for drinking, aquatic breeding, and unrestricted domestic use.

Tier 2: WQI $26 - 50$ — MODERATE

Suitable for recreational use, irrigation, and resilient fish species. Requires conventional filtration for potable use.

Tier 3: WQI $51 - 75$ — POOR / POLLUTED

Degraded ecological conditions. Requires extensive industrial effluent treatment. Direct human consumption prohibited.

Tier 4: WQI > 75 — CRITICAL / HAZARDOUS

Severe biological dead-zone. Toxic chemical shock, zero fish survival, high risk of acute environmental toxicity.

Ground-Truth Lab Validation Benchmarks (Objective 7 — Bucket Test)

SAMPLE DESCRIPTION	PH	TURBIDITY	DO	TEMP	TDS	COMPUTED WQI	STATUS
Clean Tap Water (WASA Supply)	7.3	1.2 NTU	7.8 mg/L	24.5 °C	145 ppm	**37.0**	SAFE DOMESTIC
Natural Pond Water (Dhanmondi Lake)	7.9	14.5 NTU	5.4 mg/L	27.2 °C	380 ppm	**56.0**	MODERATE AQUATIC
Industrial Effluent (Buriganga Shock)	5.1	85.0 NTU	1.8 mg/L	31.0 °C	1150 ppm	**92.5**	CRITICAL HAZARD

3. Complete Spoken Defense Script (For 5 Professors)

Follow this exact chronological script during your verbal delivery. Time budget: **6 to 7 minutes**.

Phase 1: Formal Opening & Problem Statement

00:00 – 01:00

> *"Respected Professors, honorable committee members, and evaluators — Good morning. I am Farhad Hosen, and today I am presenting our research project: 'An AI-Based Water Quality Monitoring & Pollution Prediction System'.*
>
> *In developing countries, traditional water quality testing depends on manual bucket sampling and delayed lab titration. This causes a fatal information lag: results take days, manual testing cannot detect sudden illegal midnight chemical dumps, and it is reactive rather than predictive. Our goal was to create an autonomous, cyber-physical platform that detects contamination instantly and forecasts water pollution 24 hours ahead."*

[Screen Action]: Keep the live website open: https://project-ai-rust-mu.vercel.app in full-screen mode (Press F11).

Phase 2: Architectural Stack & WQI Modeling

01:00 – 02:30

> *"Our system integrates an ESP32 microcontroller with 5 industrial-grade sensors: pH, Turbidity, Dissolved Oxygen, Temperature, and TDS. Readings stream over HTTP and WebSockets into our asynchronous FastAPI cloud backend.*
>
> *Instead of evaluating individual numbers, our backend calculates the Weighted Arithmetic Water Quality Index (WA-WQI) benchmarked against WHO and Bangladesh Department of Environment standards. Dissolved Oxygen and TDS carry our highest analytical weights. As seen on our top card, the aggregate system index currently reads 69.5, reflecting the realistic 'Poor' baseline state of Dhaka's river waterways."*

[Screen Action]: Point mouse cursor to the "SYSTEM WATER QUALITY INDEX: 69.5" card and the 5/5 Active Stations card.

Phase 3: Machine Learning Engine (Anomaly & Forecasting)

02:30 – 04:00

> *"The brain of our platform consists of two specialized machine learning models:*
> *First, an unsupervised **Isolation Forest** trained with a 10% contamination threshold. It monitors multivariate relationships to isolate abnormal chemical dumping shocks without requiring labeled failure datasets.*
> *Second, a 24-hour predictive **Random Forest Regressor**. By engineering 1-to-6 hour lag features and 3-hour rolling averages, it forecasts upcoming WQI and parameter shifts 24 hours in advance with a test R^2 of over 0.88."*

[Screen Action]: Hover over the "24-Hour AI Pollution Forecast" interactive Chart.js graph.

Phase 4: Live Interactive Demonstration (The Climax)

04:00 – 06:00

> *"Now, let us examine the live production platform. When I click 'Start Live Stream', telemetry data streams across all 5 stations in real time. On our GIS Leaflet map, stations are color-coded by environmental risk.*
>
> *Now, I will simulate an emergency industrial scenario by injecting an 'Acidic Industrial Discharge' into the Buriganga River. Instantly, pH plunges to 4.1, the Isolation Forest AI flags the reading as an Anomaly, WQI surges past 85 into Critical, and an urgent threshold incident is dispatched across the dashboard."*

> **[Screen Action]: Click [▶️ Start Live Stream], then click [Simulate Anomaly] ➔ [🧪 Acidic Industrial Discharge]. Show the alert!**

Phase 5: Conclusion & Q&A Transition

06:00 – 06:30

> *"In conclusion, this platform provides environmental agencies with an autonomous, real-time enforcement and early-warning instrument. Our codebase is open-source and ready for large-scale deployment. Thank you for your time and guidance; I am now ready for your valuable questions."*

> **[Screen Action]: Stand confidently, smile, make eye contact with each of the 5 professors, and invite their questions.**

4. Anticipated Faculty Questions & Winning Answers

Here are the 5 most probing questions university professors are guaranteed to ask, along with your scientifically grounded responses:

Q1. Why did you choose Isolation Forest instead of simple if-else threshold rules?

> **Winning Answer:** "Honorable Professor, traditional thresholding only checks individual parameters independently. In a real industrial discharge, water might have a pH of 6.7—which is technically within the safe threshold of 6.5–8.5—but if Dissolved Oxygen drops simultaneously to 1.8 mg/L and Turbidity surges to 75 NTU, a threshold system fails to trigger. **Isolation Forest** operates in a 5-dimensional feature space. It recursively partitions random sub-samples; because anomalies require significantly fewer splits to isolate, it detects complex multivariate pollution shocks with zero labeled training data."

Q2. How do you handle sensor drift, calibration errors, and bio-fouling in real river water?

> **Winning Answer:** "That is a crucial practical engineering issue. We address it through a hybrid hardware-software approach: On the hardware level, we implement 3-point buffer calibration (pH 4.01, 7.00, 10.01) and active temperature compensation using the DS18B20 digital sensor. On the software level, our ML pipeline implements 3-hour rolling averages and median filtering. This distinguishes slow, continuous calibration drift from sudden, sharp physical pollution discharges."

Q3. Why did you select Random Forest Regressor rather than Deep Learning models like LSTM or GRU?

> **Winning Answer:** "For tabular environmental time-series of this scale, **Random Forest** provides three distinct advantages: First, LSTMs require tens of thousands of continuous epochs and easily overfit on seasonal drift. Second, by engineering 1-to-6 hour lag features and rolling standard deviations, Random Forest captures non-linear temporal dynamics without gradient vanishing issues. Third, inference takes under 5 milliseconds with negligible memory overhead, making it ideal for cloud serverless deployments."

Q4. How does the Weighted Arithmetic WQI model prevent one high parameter from masking others?

> **Winning Answer:** "In the Weighted Arithmetic formulation ($WQI = \sum q_i w_i / \sum w_i$), the assigned weights are inversely proportional to the recommended limits. For critical biological survival indicators like Dissolved Oxygen and TDS, we assigned maximum weights of 0.25 each. When DO drops or TDS spikes, its individual sub-index q_i scales aggressively, overriding secondary parameters and immediately driving the composite score into the 'Poor' or 'Critical' tier."

Q5. What happens if we scale this deployment to 100+ IoT nodes along the entire river network?

> **Winning Answer:** "The platform was architected for horizontal scalability from Day 1. The FastAPI backend is completely asynchronous and non-blocking. Database transactions use indexed foreign keys and connection pooling compatible with managed PostgreSQL or Amazon RDS. On the frontend, Leaflet uses canvas-based GeoJSON clustering to render hundreds of spatial nodes at 60 frames per second."

Final Reminder for Defense Success:

Keep your voice calm, project confidence, maintain eye contact with all 5 professors, and refer directly to the live dashboard numbers. Your preparation and live demonstration will leave a lasting positive impression on the faculty board!